Memory-Induced Effects in Quantum Cosmology

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Goal

Can the evolution of the Universe depend not only on its present state, but also on its past history?

The goal of this work is to formulate a quantum cosmological framework incorporating temporal memory and to explore its theoretical and phenomenological consequences.

Why Memory?

- Locality may break down at high energies, motivating non-local theories of quantum gravity that improve the renormalizability of gravity.
- Gravitational memory phenomena motivate the search for analogous and potentially richer memory effects in quantum gravity.

Can quantum cosmological memory leave observable imprints on the Universe?

Framework

To investigate memory effects, we adopt the quantum cosmological framework developed by Claus Kiefer and collaborators.

- Canonical quantum gravity based on the Wheeler-DeWitt equation.
- Semiclassical (WKB) expansion.
- Provides a natural setting to introduce temporal memory while retaining contact with cosmological observables.

We first briefly review the Kiefer framework before introducing memory.

Semiclassical Hierarchy

Expanding the wavefunction order-by-order in m_P^2 ($\Psi_{\mathbf{k}}(q^A, \nu_{\mathbf{k}}) = \exp\left(i[m_P^2 S_0 + m_P^0 S_1 + m_P^{-2} S_2 + \dots]\right)$ yields:

- $\mathcal{O}(m_P^4)$:

$$\frac{\partial S_0}{\partial v_{\mathbf{k}}} = 0,$$

  indicating that the classical background is independent of perturbations.

- $\mathcal{O}(m_P^2)$:

$$\mathcal{G}_{AB} \frac{\partial S_0}{\partial q_A} \frac{\partial S_0}{\partial q_B} + V(q) = 0,$$

  which is the Hamilton-Jacobi equation for the classical cosmological background.

Emergence of Quantum Dynamics

At the next orders of the semiclassical expansion,

- $\mathcal{O}(m_P^0)$

$$i\frac{\partial}{\partial\eta}\psi_{\mathbf{k}}^{(0)} = \hat{H}_{\mathbf{k}}\psi_{\mathbf{k}}^{(0)},$$

  the Schrödinger equation.

- $\mathcal{O}(m_P^{-2})$
  The leading quantum-gravitational correction modifies the Schrödinger equation,

$$i\frac{\partial}{\partial\eta}\psi_{\mathbf{k}}^{(1)} = \hat{H}_{\mathbf{k}}\psi_{\mathbf{k}}^{(1)} + \mathcal{O}(m_P^{-2}),$$

This is the point where we introduce memory.

Incorporating Memory

- A direct, *ad-hoc* fractionalization of the Wheeler-DeWitt equation is not consistent.
- Instead, we introduce a causal non-local memory kernel,

$$\frac{1}{m_P^2}\int_0^{\eta} K(\eta-\eta')\frac{\partial\Psi(\eta')}{\partial\eta'}\,d\eta',$$

  which modifies the quantum-gravitational sector.
- The memory integral begins at $\eta = 0$, corresponding to the emergence of the classical Lorentzian Universe (e.g. Hartle-Hawking or Vilenkin tunneling universe).
- We choose the kernel

$$K(\eta-\eta') = -\frac{iA\,\psi_{\mathbf{k}}^{(1)}(\eta)}{\psi_{\mathbf{k}_0}^{(1)}(\eta)}\left[\frac{1}{\eta-\eta'}\right]_+.$$

Memory-Corrected Schrödinger Equation

The non-local kernel modifies the quantum-gravitational Schrödinger equation into

$$i\,{}^{C}D_{\eta}^{\alpha}\psi_{\mathbf{k}}^{(1)} = \mathcal{H}_{\mathbf{k}}\psi_{\mathbf{k}}^{(1)} + \text{Standard QG Correction} \tag{1}$$

where the fractional order is

$$\alpha = 1 - \frac{A\psi_{\mathbf{k}}^{(1)}}{m_P^2\psi_{\mathbf{k}_0}^{(1)}} \simeq 1 - \frac{A}{m_P^2}\left(\frac{k}{k_0}\right)^{3/4} = 1 - \mathcal{O}(m_P^{-2}).$$

Equivalently,

Standard QG correction + non-local memory integral.

Memory appears as a fractional time evolution, equivalent to an explicit history-dependent integral.

Implications for the Primordial Power Spectrum and CMB Temperature Anisotropies

- Memory induces a **blue-tilted quantum-gravitational correction** to the primordial scalar and tensor power spectra,

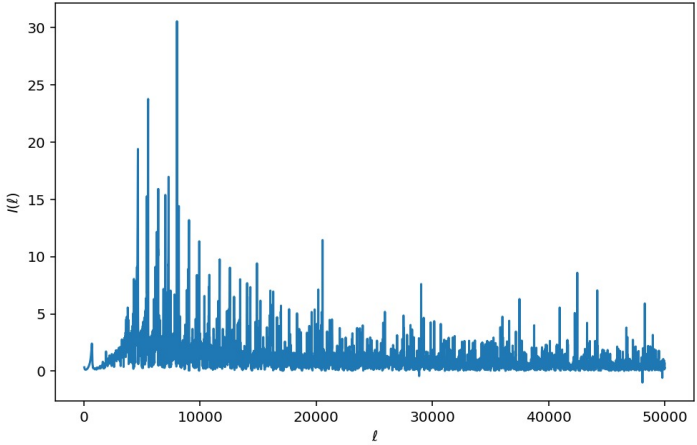
$$\mathcal{P}^{(1)}_{S,T}(k) = \mathcal{P}^{(0)}_{S,T}(k)\left[1 + \cdots + \frac{A}{2m_P^2}\left(\frac{8}{3} - 2\ln 2\right)\left(\frac{k}{k_0}\right)^{3/4}\right]. \tag{2}$$

- This modification propagates directly into the CMB temperature anisotropy spectrum,

$$\Delta\left(\frac{\ell(\ell+1)}{2\pi}C_\ell\right) = \frac{A_s A}{2m_P^2(k_0\eta_{\rm rec})^{3/4}}\frac{8-6\ln 2}{3}\,I_\ell, \qquad (\ell \gtrsim 30), \tag{3}$$

leading to the strongest deviations at high multipoles.

Numerical Behavior of the Memory Integral



The memory integral I_ℓ exhibits an enhancement toward small angular scales (large ℓ), providing a distinctive signature of non-Markovian quantum gravitational dynamics.

Detectability of Memory Effects

Comparing the memory-induced correction with the standard CMB temperature anisotropy spectrum suggests that observable effects arise for

$$\frac{A}{m_P^2} \gtrsim 10^{-3}\text{–}10^{-4}.$$

Even modest memory couplings can produce appreciable corrections to the CMB temperature anisotropy spectrum at large ℓ.

Higher-Order Correlation Functions

The memory kernel modifies not only the primordial power spectrum but also higher-order correlation functions.

- The induced bispectrum inherits the characteristic blue-tilted memory correction.
- Scale-dependent corrections arise in the squeezed-limit non-Gaussianity parameter f_{NL}.
- Equilateral and folded bispectra exhibit characteristic $k^{3/4}$ and logarithmic scale dependence.

Thank You!